

Clinical Atlas Systematic Image Research and GIF Pipeline

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Status: Implementation and editorial research plan

1. Purpose

The current clinical gallery should be transformed from a collection of photographs into a coherent visual education system.

The objective is not to add more images randomly. The objective is to make every visual answer a clear question:

- What is being explained?
- Why is this visual necessary?
- What should the learner notice?
- What should the learner do next?
- What can the visual not prove?

The final product should combine:

- Premium editorial composition.
- Meaningful SVG diagrams.
- Short explanatory animations.
- Carefully selected GIF-like loops where motion teaches a process.
- Clinical photography only where justified.
- Clear captions and descriptions.
- Source and rights tracking.
- Learner and facilitator separation.

2. Important clinical and legal guardrail

Do not treat a web search result, downloaded image, PDF, CDC slide, or stock image as automatically reusable or clinically authoritative.

NACO's official STI/RTI services page describes counselling, syndromic colour-coded drug kits, condoms, referrals for HIV and syphilis screening, and referrals to STI laboratories. It also

provides a 2023 technical specification resource for kits, dual-test kits, and condoms. These official resources should be the starting point for Indian programme context. [web:55][web:57]

The older NACO report lists seven colour-coded kits and their historical syndromic associations, but its underlying guideline reference is older. Current mapping and treatment content must be checked against the latest applicable NACO/MoHFW guidance before publication. [web:56]

CDC's STI Clinical Slides are designed for healthcare providers and health educators and may not be appropriate for every audience. [web:60] CDC also states that some government images are public domain while other images are third-party licensed, so each asset must be checked individually and credited correctly. [web:66]

3. Visual content strategy

Use a four-layer visual hierarchy.

Layer 1 — Concept illustrations

Default for learners:

- Custom SVG characters.
- Pathway diagrams.
- Anatomy-neutral illustrations.
- Clinic journey diagrams.
- Colour-coded care objects.
- Simple motion graphics.

Purpose: explain concepts without fear, shame, or diagnostic interpretation.

Layer 2 — Animated explanations

Use short loops or SVG timelines for:

- How transmission pathways can differ.
- How a test journey works.
- How a page or kit is opened.
- How a hidden symptom concept is revealed.
- How partner care reduces repeated infection risk.
- How a clinic referral pathway works.

Purpose: show a process or relationship that a still image cannot express.

Layer 3 — Carefully selected educational photographs

Use only when a photograph provides a learning value that an illustration cannot provide.

Requirements:

- Rights cleared.
- Clinical interpretation verified.
- Audience suitability reviewed.
- Content shield available.
- Non-diagnostic wording.
- Accessible alternative.
- Source and photographer credited.

Layer 4 — Restricted professional atlas

For trained health workers only where justified:

- Higher-sensitivity imagery.
- Detailed clinical captions.
- Protocol references.
- Review date.
- Authenticated access.
- Audit trail.

4. Visual decision tree

Before adding an image, ask:

```
Can an SVG explain this clearly?
-> Yes: use SVG.
-> No: can a short animation explain it?
    -> Yes: use SVG/CSS/GIF-like motion.
    -> No: is a photograph clinically necessary?
        -> Yes: verify rights, consent, clinical meaning, and audience.
        -> No: do not add it.
```

5. Systematic web research workflow

5.1 Build a source allowlist

Search first within:

- NACO/MoHFW official sites.
- WHO official sites.

- CDC official sites.
- Government public-health image libraries.
- Licensed medical publishers.
- Reputable academic or institutional collections.

Do not begin with generic image search for final assets.

5.2 Search query groups

Use focused queries rather than broad image scraping.

Programme and care

```
site:naco.mohfw.gov.in STI RTI Suraksha clinic
site:naco.mohfw.gov.in STI colour coded kits
site:naco.mohfw.gov.in syndromic case management STI
```

Public-health illustrations

```
site:who.int STI prevention testing illustration
site:cdc.gov STI prevention testing image
site:cdc.gov/phil STI clinical image
```

Disease-specific research

```
site:cdc.gov/phil primary syphilis clinical image
site:cdc.gov/phil secondary syphilis palms
site:cdc.gov/phil genital herpes clinical image
site:cdc.gov/phil gonorrhea clinical image
site:cdc.gov/phil genital warts clinical image
```

Process and care illustrations

```
WHO STI testing clinic illustration
public health clinic counselling illustration
sexual health testing journey illustration
```

5.3 Record every search result

Do not download directly into public/.

Create a research record:

```
{
  "researchId": "research-0001",
  "query": "site:cdc.gov/phil secondary syphilis palms",
```

```
"sourceUrl": "https://example.org/image",
"sourceType": "official-image-library",
"assetId": "asset-0001",
"candidateUse": "clinical-reference",
"rightsStatus": "pending",
"clinicalStatus": "pending",
"audience": "restricted",
"notes": "Requires individual rights and sensitivity review"
}
```

6. Web research pipeline

```
Search
-> Shortlist
-> Open source page
-> Verify creator and license
-> Verify image identity
-> Record source metadata
-> Download to quarantine
-> Hash original
-> Generate contact sheet
-> Clinical review
-> Rights review
-> Audience review
-> Accessibility description
-> Create derivative
-> Approve or reject
-> Publish into correct bundle
```

7. Asset registry

Create content/clinical/asset-registry.json with:

```
assetId
sourceUrl
sourceTitle
sourceOrganization
creator
photographer
originalFilename
originalChecksum
localQuarantinePath
derivativePaths
licenseType
licenseText
attributionText
permissionEvidence
consentStatus
clinicalInterpretation
clinicalReviewer
clinicalStatus
sensitivityLevel
allowedAudience
```

```
allowedLocales
altText
caption
reviewDate
nextReviewDate
publicationStatus
```

8. Rights classification

Green — cleared

- Explicit permission or confirmed public-domain status.
- Attribution requirements recorded.
- Clinical and audience review complete.

Amber — review required

- Source organization appears trustworthy but license is unclear.
- Image may be reusable under conditions.
- Permission request pending.

Red — rejected

- License unknown.
- Third-party restrictions.
- Consent unavailable.
- Image identity uncertain.
- Clinical interpretation disputed.
- Audience risk too high.

Do not ship amber or red assets.

9. CDC image handling

The CDC provides a Public Health Image Library and states that many images are royalty-free for certain uses with appropriate citation, while other images may be third-party licensed. Check the individual image record rather than assuming all CDC-hosted images have the same rights. [web:64][web:66]

For every CDC/PHIL image:

- Capture the PHIL ID.
- Record content provider.
- Record photographer if listed.
- Record copyright restriction.

- Record attribution.
- Record the source page.
- Record whether web linking is required.
- Confirm whether republication is allowed.

10. NACO image and branding handling

Use NACO official pages and technical specifications for programme facts and kit design references. Do not reproduce NACO logos or government marks as branding without permission. [web:55][web:57][web:59]

For kit visuals:

- Prefer custom 3D/CSS/SVG representations.
- Do not imply that the application dispenses medicines.
- Do not show a kit as a personal prescription.
- Label the visual as an educational representation if it is not an official package image.
- Verify kit colour and contents against current official guidance.

11. GIF and animation policy

GIFs should not be added just because they are visually attractive. Use motion only when it communicates:

- Sequence.
- Direction.
- Change.
- Comparison.
- Cause and effect.
- Attention to a relevant visual target.

11.1 Prefer animated SVG or CSS over GIF

Use animated SVG/CSS for:

- Small diagrams.
- Page and kit interactions.
- Highlighting.
- Path drawing.
- Simple explanatory loops.

Advantages:

- Crisp at all sizes.

- Better control of timing.
- Smaller for simple shapes.
- Easier target IDs.
- Easier reduced-motion behaviour.
- Better integration with narration.

11.2 Use GIF/WebP animation for

- Short pre-rendered artistic loops.
- Complex texture motion.
- Approved non-interactive decorative sequences.
- Fallback on systems where interactive animation is not available.

Requirements:

- Keep loops short.
- Avoid flashing.
- Provide static poster frame.
- Add alt text.
- Do not make essential meaning depend on loop timing.
- Respect reduced motion.

11.3 Animation metadata

```
export type AnimationAsset = {
  id: string
  type: "svg" | "css" | "webp" | "gif" | "lottie"
  purpose: "explanation" | "interaction" | "decoration"
  durationMs: number
  loop: boolean
  staticFallback: string
  reducedMotionBehaviour: "freeze" | "fade" | "replace"
  altText: string
  sourceStatus: "custom" | "licensed" | "public-domain"
}
```

12. Premium visual direction

12.1 Replace clinical grid with visual storytelling

Instead of:

```
Image | Image | Image
Label | Label | Label
```


Use:

Question

- > Illustrated body/care context
- > Focused visual detail
- > Plain-language explanation
- > Professional-care action

12.2 Clinical atlas composition

Each atlas entry should include:

- Neutral title.
- “What a health worker may notice” wording.
- One approved visual at a time.
- Male/anatomical presentation and vulva/cervical presentation where clinically meaningful.
- Extragenital presentation only when relevant.
- Source/citation panel.
- Non-diagnostic disclaimer.
- Illustrated alternative.
- “Find care” action.

12.3 Do not overload a spread

One spread should not show nine diseases, multiple photographs, all regimens, and all follow-up schedules simultaneously.

Use progressive disclosure:

1. Concept page.
2. Syndrome page.
3. One selected atlas entry.
4. Optional professional detail.
5. Referral/action page.

13. Definition-writing standard

Current definitions need to become more meaningful and less like labels.

Use a three-layer definition:

Layer 1 — Plain-language meaning

What the reader may notice or feel, without diagnosing.

Layer 2 — Why professional care matters

Explain that similar symptoms may have different causes and that a health worker may use questions, examination, and tests.

Layer 3 — Action

Tell the user what safe step to consider.

Template:

What this page explains:

A symptom or body change that a health worker may assess.

What it does not mean:

A picture cannot confirm a diagnosis.

What to do:

Speak with a qualified health worker about testing and care.

14. Recommended visual modules

Module A — Symptom language cards

Use illustrated cards for:

- Discharge.
- Sores or ulcers.
- Blisters.
- Itching.
- Pain.
- Swelling.
- No visible signs.

Each card explains:

- What the word means.
- Why it can have different causes.
- When to seek professional guidance.

Module B — Transmission pathway animation

Use abstract nodes and pathways rather than explicit imagery:

- Contact pathway.
- Barrier/protection pathway.
- Testing pathway.
- Professional-care pathway.

Module C — Clinic journey animation

Animated sequence:

```
Arrive
-> Talk privately
-> Health worker asks questions
-> Assessment/testing
-> Guidance
-> Follow-up if advised
```

Module D — NACO kit cabinet

Use custom SVG/3D kit boxes with:

- Colour.
- Number.
- Syndrome category.
- Learner-safe explanation.
- Facilitator reference link.

Module E — Clinical atlas reveal

Use a sensitive-content shield and illustrated alternative before any photograph.

15. Recommended image-search manifest

Create content/visual-research/visual-search-manifest.json:

```
{
  "collections": [
    {
      "id": "clinic-journey",
      "purpose": "Explain what happens during a care visit",
      "preferredFormats": ["svg", "illustration", "animation"],
      "searchQueries": [
        "site:who.int STI testing clinic illustration",
        "site:naco.mohfw.gov.in Suraksha clinic",
        "public health counselling clinic illustration"
      ]
    }
  ]
}
```

```

    ],
    "fallback": "commission-custom-svg",
    "status": "planned"
  },
  {
    "id": "clinical-syphilis",
    "purpose": "Restricted clinical reference only",
    "preferredFormats": ["licensed-photo", "illustration"],
    "searchQueries": [
      "site:cdc.gov/phil primary syphilis",
      "site:cdc.gov/phil secondary syphilis palms"
    ],
    "fallback": "custom-clinical-illustration",
    "audience": "facilitator",
    "status": "rights-and-clinical-review-required"
  }
]
}

```

16. Proposed implementation files

```

content/visual-research/
├─ visual-search-manifest.json
├─ search-results.json
├─ asset-shortlist.json
└─ rejected-assets.json

content/clinical/
├─ asset-registry.json
├─ claim-registry.json
├─ protocol-registry.json
└─ clinical-atlas-registry.json

scripts/
├─ research_image_sources.py
├─ download_quarantine_assets.py
├─ generate_contact_sheet.py
├─ validate_asset_rights.py
├─ validate_clinical_metadata.py
├─ convert_approved_assets.py
└─ generate_asset_report.py

components/visuals/
├─ AnimatedSvgScene.tsx
├─ StaticVisualFallback.tsx
├─ VisualTarget.tsx
├─ VisualCaption.tsx
└─ MotionPreference.tsx

components/clinical/
├─ ClinicalContentShield.tsx
├─ IllustratedAlternative.tsx
├─ ClinicalPhotoGallery.tsx
└─ ClinicalZoomModal.tsx

```

```
|— SourceCitationPanel.tsx
|— ClinicalAtlasEntry.tsx
```

17. Search automation requirements

A research script may:

- Read a pre-approved URL list.
- Fetch metadata.
- Save source URLs.
- Save thumbnails to quarantine.
- Calculate checksums.
- Create contact sheets.
- Generate review records.

It must not:

- Crawl the open web indiscriminately.
- Download unknown images directly into production.
- Assume search snippets are evidence.
- Infer disease identity automatically.
- Remove watermarks.
- Bypass access controls.
- Publish assets without review.

18. Quality scoring for candidate visuals

Score each candidate from 0 to 5:

```
Educational necessity:
Clinical identity confidence:
Rights clarity:
Audience suitability:
Accessibility potential:
Resolution:
Cultural appropriateness:
Non-diagnostic safety:
```

Reject any candidate with:

- Rights clarity below 4.
- Clinical identity confidence below 4 for a clinical claim.
- Non-diagnostic safety below 4.
- Audience suitability below 4.

19. Verification plan

Automated

- Registry schema validation.
- Source URL validation.
- Rights status validation.
- Missing alt-text detection.
- Duplicate checksum detection.
- Static fallback detection.
- Public/restricted bundle separation.
- Animation metadata validation.
- Reduced-motion fallback validation.

Manual

- Open each source page.
- Verify rights statement.
- Confirm creator and attribution.
- Confirm image identity with clinical reviewer.
- Review sensitivity and audience.
- Check crop and colour changes.
- Review accessibility description.
- Test mobile and desktop layout.

User comprehension

Ask:

- “What does this visual explain?”
- “Can this image tell you what condition you have?”
- “What would you do next?”
- “Would you prefer this illustration or photograph?”

20. Success criteria

The image system is successful when:

- Every image has a documented purpose.
- Every image has verified rights.
- Every clinical image has clinical review.

- Learners understand that photographs do not diagnose.
- Illustrations carry the main learning burden.
- GIFs/animations explain processes rather than decorate.
- The page remains premium without becoming crowded.
- The public bundle excludes unnecessary restricted images.
- The content team can find, review, approve, replace, and withdraw assets systematically.

21. Recommended immediate tasks

1. Freeze current clinical photography publication.
2. Create the asset registry and quarantine folders.
3. Record every existing image's source, page, rights, and status.
4. Obtain current NACO/MoHFW clinical review.
5. Build the illustrated symptom and clinic-journey system.
6. Search official sources using the query manifest.
7. Shortlist only rights-clear or reviewable assets.
8. Create three meaningful animated SVG scenes.
9. Add the sensitive-content shield and illustrated alternative.
10. Test comprehension before adding more photographs.